The sealed statements of FermionicLean

These 79 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

Covariance matrices

A system of n fermionic modes $j = 0, \dots, n-1$ has $2n$ Majorana labels: x_j and y_j label the two Majorana operators of mode j. The label set is the disjoint union of two copies of $\{0, \dots, n-1\}$, the first copy carrying the x labels, so a matrix on the labels is written in 2×2 block form with the x labels first. A covariance matrix is a real matrix Γ on the labels. The covariance of a state ρ with Majorana operators c_a is

$$\Gamma_{ab} = \operatorname{Re} \tfrac{i}{2} \operatorname{tr}(\rho[c_a, c_b])$$

(second part); with this sign the vacuum has covariance $-\Omega_n$, where

$$\Omega_n = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

so $\Gamma_{x_j y_j} = -1$ in the vacuum. A real matrix Γ on a finite index set is admissible when $\Gamma^T = -\Gamma$ and $1 - \Gamma^T\Gamma$ is positive semidefinite, and pure when $\Gamma^T = -\Gamma$ and $\Gamma^2 = -1$. A complex matrix X is positive semidefinite when it is Hermitian and $\bar{v}^T X v \geq 0$ for every complex vector v; for real matrices the vectors are real. A real matrix O is orthogonal when $OO^T = 1$.

1. The definitions. The Majorana labels of n modes are two copies of $\{0, \dots, n-1\}$, the x labels first. A real matrix Γ on them is admissible exactly when $\Gamma^T = -\Gamma$ and $1 - \Gamma^T\Gamma$ is positive semidefinite, and pure exactly when $\Gamma^T = -\Gamma$ and $\Gamma\Gamma = -1$. The same conditions define admissibility and purity of a real matrix Γ' on an arbitrary finite index set ι.

`Sealed.admissible_def`

2. Equivalent forms of admissibility. For a real antisymmetric matrix Γ on the Majorana labels of n modes, the following are equivalent: Γ is admissible; $\|\Gamma v\|^2 \leq \|v\|^2$ for every real vector v; the complex matrix $1 + i\Gamma$ is positive semidefinite.

`Sealed.admissible_forms`

3. A pure covariance matrix Γ on the Majorana labels of n modes is orthogonal, $\Gamma^T\Gamma = 1$, and admissible.

`Sealed.pure_orthogonal`

4. Orthogonal changes of Majorana basis. Let O be a real orthogonal matrix on the Majorana labels of n modes ($OO^T = 1$). If Γ is admissible, so is $O\Gamma O^T$; if Γ is pure, so is $O\Gamma O^T$.

`Sealed.orthogonal_congruence`

5. Relabelling and block-diagonal sums. Let $e : \iota \to \kappa$ be a bijection of finite index sets and Γ a real matrix on κ; the relabelled matrix $(\Gamma_{e(a)e(b)})_{a,b\in\iota}$ is admissible exactly when Γ is, and pure exactly when Γ is. Let Δ_s, $s \in \sigma$, be real matrices on a finite set ι, with σ finite, and $\bigoplus_s \Delta_s$ the block-diagonal matrix on $\iota \times \sigma$ with entry $\delta_{st}(\Delta_s)_{ab}$ at $((a,s),(b,t))$; it is admissible exactly when every Δ_s is, and pure exactly when every Δ_s is.
`Sealed.relabel_block_diagonal`

6. Restriction to a subset of modes. Let Γ be an admissible covariance matrix of n modes and $e : \{0,\dots,m-1\} \to \{0,\dots,n-1\}$ injective. The restriction of Γ to the labels $x_{e(j)}, y_{e(j)}$, $j = 0,\dots,m-1$, in this order, is an admissible covariance matrix of m modes.
`Sealed.mode_restriction`

7. Independent sums. Let A be a real matrix on the Majorana labels of n modes and B one on those of m modes. Their independent sum $A \oplus B$ is a real matrix on the labels of $n+m$ modes, the modes of A first: the map σ with $\sigma(x_j^A) = x_j$, $\sigma(y_j^A) = y_j$, $\sigma(x_j^B) = x_{n+j}$, $\sigma(y_j^B) = y_{n+j}$ is a bijection from the labels of A followed by those of B onto the labels of $n+m$ modes, and $(A \oplus B)_{\sigma(u)\sigma(v)}$ is the (u,v) entry of

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}.$$

The sum is admissible exactly when A and B are, and pure exactly when A and B are.
`Sealed.direct_sum`

8. One mode. Every real antisymmetric matrix Γ on the two labels of one mode is $\mu\omega$ for some real μ, where

$$\omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix};$$

and for real λ, $\lambda\omega$ is admissible exactly when $|\lambda| \le 1$ and pure exactly when $|\lambda| = 1$.
`Sealed.one_mode`

9. The standard pure covariances. On the Majorana labels of n modes,

$$\Omega_n = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

and $-\Omega_n$ (the vacuum covariance in the sign convention above) are pure.
`Sealed.standard_pure`

Contraction and the bond

Let ι (retained labels) and κ (eliminated labels) be finite index sets, Γ a real matrix on $\iota \sqcup \kappa$ with blocks $A = \Gamma_{\iota\iota}$, $B = \Gamma_{\iota\kappa}$, $C = \Gamma_{\kappa\iota}$, $D = \Gamma_{\kappa\kappa}$, and β a real matrix on κ. The contraction of Γ with β is the matrix

$$\Gamma \star \beta = A + B\beta(1 - D\beta)^{-1}C$$

on ι. It is meant under the contraction condition $\det(1 - D\beta) \neq 0$; the inverse is the matrix inverse, which is 0 for a singular matrix. For a finite set L of modes, the bond is a covariance matrix of two matched copies L, L' of L, on the labels ordered $x_L, x_{L'}, y_L, y_{L'}$; it correlates x_a with $x_{a'}$ and y_a with $y_{a'}$ for every $a \in L$.

10. The definition of the contraction:

$$\Gamma \star \beta = A + B\beta(1 - D\beta)^{-1}C$$

in the blocks of Γ, and the contraction condition is $\det(1 - D\beta) \neq 0$.
`Sealed.contract_def`

11. Contraction preserves antisymmetry and admissibility. Under the contraction condition: if Γ and β are antisymmetric, so is $\Gamma \star \beta$; if Γ and β are admissible, so is $\Gamma \star \beta$.
`Sealed.contraction_admissible`

12. Contraction with a pure β. If β is pure, the contraction condition holds exactly when $\det(D + \beta) \neq 0$, and for antisymmetric Γ satisfying it,

$$\Gamma \star \beta = A + B(D + \beta)^{-1}B^T.$$

`Sealed.contraction_pure_bond`

13. Contraction of pure covariances is pure. Let Γ be a pure matrix on the Majorana labels of n retained modes followed by those of m eliminated modes, and β a pure covariance matrix of the m eliminated modes, with the contraction condition. Then $\Gamma \star \beta$ is a pure covariance matrix of the n retained modes.
`Sealed.contraction_pure`

14. The bond. For a finite set L, the labels of the bond are four copies of L, ordered $x_L, x_{L'}, y_L, y_{L'}$; the bond is

$$\begin{pmatrix} \omega & 0 \\ 0 & -\omega \end{pmatrix}$$

with

$$\omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

on (L, L'), so $\Gamma_{x_a x_{a'}} = 1$ and $\Gamma_{y_a y_{a'}} = -1$ for $a \in L$. The bond is pure.
`Sealed.bond_pure`

15. Entanglement swapping through a bond. Take two bonds, on (A_1, B_1) and on (A_2, B_2), each mode set a copy of L and each bond on its labels (x_A, x_B, y_A, y_B), and form their block-diagonal sum. Read it on the retained labels $(x_{A_1}, x_{B_2}, y_{A_1}, y_{B_2})$ followed by the eliminated labels $(x_{B_1}, x_{A_2}, y_{B_1}, y_{A_2})$, through the map τ sending each of these labels to the label of the same name in the sum. The result is

$$\begin{pmatrix} 0 & X \\ -X^T & 0 \end{pmatrix}$$

with $X = \mathrm{diag}(1, -1, -1, 1)$ in blocks of size $|L|$, and its contraction with the bond on (B_1, A_2) is the bond on (A_1, B_2).
`Sealed.teleportation`

Periodic chains and three sites

A periodic chain has N sites $0, \ldots, N-1$, read modulo N; $i+1$ and $i-1$ are the cyclic successor and predecessor. Each site carries M left virtual modes A, M right virtual modes B and P physical modes C (each numbered from 0); its Majorana labels are ordered $x_A, x_B, x_C, y_A, y_B, y_C$. The chain data are real matrices Γ_i on the labels of site i. The retained labels of the chain are

the physical labels $x_{(i,c)}, y_{(i,c)}$ of all sites, x first. The eliminated labels are the bond labels $x_L, x_{L'}, y_L, y_{L'}$ with

$$L = L' = \{\text{sites}\} \times \{\text{virtual modes}\} :$$

the label (i, a) of L is the mode A_a of site i, and the label (i, a) of L' is the mode B_a of site $i - 1$, so that bond i joins site $i - 1$ to site i. The periodic input is the block-diagonal sum of the Γ_i on these labels, the periodic bond is the bond on L, and the periodic covariance Γ_N is the contraction of the periodic input with the periodic bond, a matrix on the physical labels.

16. The label sets of a chain: the labels of a site are (A, B, C) for x, then (A, B, C) for y; the retained labels are the pairs (site, physical mode) for x, then for y; the eliminated labels are the bond labels on the pairs (site, virtual mode).

`Sealed.chain_labels`

17. The definition of the periodic chain. The periodic input of chain data Γ_i is the block-diagonal matrix $\bigoplus_i \Gamma_i$ on (site label, site) relabelled by σ, which sends $x_{(i,c)} \mapsto (x_{C_c}, i)$, $y_{(i,c)} \mapsto (y_{C_c}, i)$, the L labels $x_{(i,a)} \mapsto (x_{A_a}, i)$, $y_{(i,a)} \mapsto (y_{A_a}, i)$, and the L' labels $x_{(i,a)} \mapsto (x_{B_a}, i - 1)$, $y_{(i,a)} \mapsto (y_{B_a}, i - 1)$. The periodic bond is the bond on L, and the periodic covariance is the contraction of the periodic input with the periodic bond.

`Sealed.periodic_def`

18. Admissibility of the chain. The periodic input is admissible exactly when every Γ_i is, and pure exactly when every Γ_i is. If every Γ_i is admissible and the contraction condition holds for the periodic input and the periodic bond, the periodic covariance Γ_N is admissible.

`Sealed.periodic_admissible`

19. Purity of the chain. If every Γ_i is pure and the contraction condition holds, the periodic covariance Γ_N is pure.

`Sealed.periodic_pure`

20. A three-site chain, evaluated exactly. Take $N = 3$ sites, $M = P = 1$, and at every site the matrix G that, on the labels ordered $(x_A, x_B, x_C, y_A, y_B, y_C)$, is

$$\tfrac{1}{5}\begin{pmatrix} 0 & 3 & -4 & 0 & 0 & 0 \\ -3 & 0 & 0 & 0 & 0 & -4 \\ 4 & 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 4 & 3 & 0 & 0 & 0 \end{pmatrix}.$$

Then G is pure, the contraction condition holds with $\det(1 - D\beta_3) = (304/125)^2$ (D the block of the input on the eliminated labels, β_3 the bond), and the periodic covariance, on the physical labels ordered $(x_{C_0}, x_{C_1}, x_{C_2}, y_{C_0}, y_{C_1}, y_{C_2})$, is

$$\tfrac{1}{19}\begin{pmatrix} 0 & 0 & 0 & -15 & 6 & -10 \\ 0 & 0 & 0 & -10 & -15 & 6 \\ 0 & 0 & 0 & 6 & -10 & -15 \\ 15 & 10 & -6 & 0 & 0 & 0 \\ -6 & 15 & 10 & 0 & 0 & 0 \\ 10 & -6 & 15 & 0 & 0 & 0 \end{pmatrix}.$$

`Sealed.three_site`

Momentum space

A homogeneous chain has the same matrix Γ on the labels of every site, and $N \geq 1$ sites. For sites i and momenta k in $\{0, \ldots, N-1\}$ the character is $\chi_k(i) = e^{-2\pi iki/N}$ and the phase of one step is $\omega_k = e^{-2\pi ik/N}$; the momentum $-k$ is $N-k$ modulo N. The twist t_k is the function on the labels of a site that is ω_k on the labels x_B, y_B and 1 on all others. The twisted local matrix

$$T_k = \mathrm{diag}(t_k)\,\Gamma\,\mathrm{diag}(t_k)^*$$

is a complex matrix, written in the split of the site labels into the physical labels (x_C, y_C) and the virtual labels

$$(x_L, x_{L'}, y_L, y_{L'}) = (x_A, x_B, y_A, y_B).$$

The momentum-space covariance $\Sigma(k)$ is the contraction of T_k with the bond β_M of one site's M virtual modes, computed over $\mathbb{C}$ with the complex matrix inverse; it is a complex matrix on the physical labels (x_C, y_C) of one site. A physical label u of a site, placed at site i, is written (i, u); Γ_N is the periodic covariance of the homogeneous chain. For a complex matrix, X^* is the conjugate transpose and $\bar{X}$ the entrywise conjugate.

21. The definition of the momentum-space covariance. With $\omega_k = e^{-2\pi ik/N}$, the twist t_k equal to ω_k on x_B, y_B and 1 elsewhere, and s the split of the site labels sending the physical labels x_c, y_c to x_{C_c}, y_{C_c} and the virtual labels $x_L, x_{L'}, y_L, y_{L'}$ to x_A, x_B, y_A, y_B: the twisted local matrix is

$$T_k = (\mathrm{diag}(t_k)\,\Gamma\,\mathrm{diag}(t_k)^*)_{s(u)s(v)},$$

the one-site bond is the bond on M modes regarded as a complex matrix, and

$$\Sigma(k) = A_k + B_k\beta_M(1 - D_k\beta_M)^{-1}C_k$$

in the blocks A_k, B_k, C_k, D_k of T_k (physical, then virtual).

`Sealed.momentum_def`

22. The determinant factorizes over momenta. For a homogeneous chain,

$$\det(1 - D\beta_N) = \prod_k \det(1 - D_k\beta_M),$$

where D is the virtual block of the periodic input, β_N the periodic bond, and D_k the virtual block of T_k. In particular, under the contraction condition of the chain every $\det(1 - D_k\beta_M)$ is nonzero.

`Sealed.momentum_det`

23. Diagonalization by the Fourier transform. For a homogeneous chain with the contraction condition, momenta k, l and physical labels a, b of a site,

$$\frac{1}{N}\sum_{i,j} \chi_k(i)\,(\Gamma_N)_{(i,a)(j,b)}\,\overline{\chi_l(j)} = \delta_{kl}\,\Sigma(k)_{ab}.$$

`Sealed.fourier_diagonal`

24. Reconstruction and translation invariance. For a homogeneous chain with the contraction condition, sites i, j and physical labels a, b of a site,

$$(\Gamma_N)_{(i,a)(j,b)} = \frac{1}{N}\sum_k \overline{\chi_k(i)}\,\Sigma(k)_{ab}\,\chi_k(j),$$

and

$$(\Gamma_N)_{(i+1,a)(j+1,b)} = (\Gamma_N)_{(i,a)(j,b)}.$$

`Sealed.fourier_reconstruct`

25. Symmetries of the momentum-space covariance. For a homogeneous chain and every momentum k: $\Sigma(-k) = \overline{\Sigma(k)}$. If Γ is antisymmetric and the contraction condition holds, $\Sigma(k)^* = -\Sigma(k)$. If Γ is admissible and the contraction condition holds, $1 - \Sigma(k)^*\Sigma(k)$ is positive semidefinite.

`Sealed.momentum_symmetries`

Normal form and rotations

For real numbers $\varepsilon_0, \ldots, \varepsilon_{n-1}$, the block form $D(\varepsilon)$ is the real matrix on the Majorana labels of n modes with $D(\varepsilon)_{x_j y_j} = \varepsilon_j$, $D(\varepsilon)_{y_j x_j} = -\varepsilon_j$ and all other entries 0. A plane rotation on a finite index set ι acts as the rotation by the angle with cosine c and sine s in the plane of two distinct labels a, b and as the identity elsewhere. The exponential of a square matrix is the sum of its exponential series.

26. The block form:

$$D(\varepsilon) = \begin{pmatrix} 0 & \operatorname{diag}(\varepsilon) \\ -\operatorname{diag}(\varepsilon) & 0 \end{pmatrix}.$$

It is admissible exactly when every $|\varepsilon_j| \le 1$, and pure exactly when every $|\varepsilon_j| = 1$.

`Sealed.block_form`

27. The normal form. Every real antisymmetric matrix Γ on the Majorana labels of n modes is $\Gamma = OD(\varepsilon)O^T$ with O orthogonal and every $\varepsilon_j \ge 0$; and also $\Gamma = O'D(\varepsilon')O'^T$ with O' orthogonal of determinant 1.

`Sealed.normal_form`

28. The values of the normal form. If $\Gamma = OD(\varepsilon)O^T$ with O orthogonal, then Γ is admissible exactly when every $|\varepsilon_j| \le 1$, and pure exactly when every $|\varepsilon_j| = 1$.

`Sealed.normal_form_values`

29. The exponential of a block form. For real $\theta_0, \ldots, \theta_{n-1}$,

$$e^{D(\theta)} = \begin{pmatrix} \operatorname{diag}(\cos\theta) & \operatorname{diag}(\sin\theta) \\ -\operatorname{diag}(\sin\theta) & \operatorname{diag}(\cos\theta) \end{pmatrix},$$

and this matrix R is orthogonal, $RR^T = 1$.

`Sealed.block_exponential`

30. Rotations generate the special orthogonal group. For labels $a \ne b$ of a finite index set ι, the plane rotation $R_{ab}(c, s)$ has rows a and b equal to $(c, -s)$ and (s, c) on the columns (a, b) and zero elsewhere, and every other row equal to that of the identity (for $a = b$ the defining formula gives the identity with c at (a, a); the products below use only $a \ne b$). A real matrix O on ι is orthogonal with $\det O = 1$ exactly when it is a finite product (possibly empty) of plane rotations $R_{ab}(c, s)$ with $a \ne b$ and $c^2 + s^2 = 1$.

`Sealed.rotation_products`

Examples: product chains and the Kitaev chain

A chain without virtual modes ($M = 0$) has no bonds. The Kitaev chain at its solvable point has one physical and one virtual mode per site ($M = P = 1$); the labels of a site are written $x_A, x_B, x_C, y_A, y_B, y_C$, and those of the chain x_i, y_i for the physical mode of site i. For labels u, v, E_{uv} is the matrix with entry 1 at (u, v) and 0 elsewhere.

31. Chains without virtual modes are products. For chain data Γ_i with $M = 0$, the contraction condition holds; the periodic covariance couples only labels of the same site,

$$(\Gamma_N)_{(i,a)(j,b)} = \delta_{ij}(\Gamma_i)_{C_a C_b}$$

for physical labels a, b, where C_a is the site label x_{C_c} for $a = x_c$ and y_{C_c} for $a = y_c$; and Γ_N is admissible exactly when every Γ_i is, pure exactly when every Γ_i is.
`Sealed.product_chain`

32. The Kitaev on-site covariance

$$K = E_{x_C x_A} - E_{x_A x_C} + E_{y_C x_B} - E_{x_B y_C}$$

correlates x_C with x_A and y_C with x_B. It is admissible and not pure.
`Sealed.kitaev_local`

33. The Kitaev chain. The matrix K_N on the physical labels of N sites has $(K_N)_{y_i x_j} = 1$ and $(K_N)_{x_j y_i} = -1$ when $j = i+1$, and all other entries 0. For $N \geq 1$, the chain with K at every site satisfies the contraction condition, its periodic covariance is K_N, and K_N is pure.
`Sealed.kitaev_chain`

34. The Kitaev chain in momentum space. For $N \geq 1$ and every momentum k, the momentum-space covariance of the Kitaev on-site covariance K is

$$\Sigma(k) = \begin{pmatrix} 0 & -\omega_k \\ \overline{\omega_k} & 0 \end{pmatrix}$$

on (x_C, y_C), with $\omega_k = e^{-2\pi i k/N}$.
`Sealed.kitaev_momentum`

35. A pure completion of the Kitaev on-site covariance. The matrix

$$K' = K - E_{y_A y_B} + E_{y_B y_A}$$

is pure. For $N \geq 1$, the chain with K' at every site has $\det(1 - D\beta_N) = (1 - (-1)^N)^2$, where D is the virtual block of the periodic input and β_N the periodic bond, so the contraction condition fails for even N; for odd N it holds and the periodic covariance is the Kitaev chain K_N.
`Sealed.kitaev_completion`

Entanglement rank

For a periodic chain and a set S of sites, R_S is the set of physical labels of the sites in S and R_{S^c} that of the sites outside S. For a matrix X on the physical labels, the cross block is $C_S = X_{R_S R_{S^c}}$ and the block covariance is $\Gamma_S = X_{R_S R_S}$. Bond i joins site $i-1$ to site i and is cut by S when exactly one of the two sites lies in S; $b(S)$ is the number of cut bonds. The rank of a matrix is the dimension of its column space.

36. The definitions of the cut. The site of the physical labels $x_{(i,c)}, y_{(i,c)}$ is i; R_S and R_{S^c} are the physical labels whose site is in S, respectively not in S; the cross block C_S and the block covariance Γ_S are the corresponding submatrices of X; and the cut bonds are the sites i such that exactly one of i and $i-1$ lies in S.
`Sealed.cut_def`

37. The cross-block rank bound. For chain data Γ_i with the contraction condition and every set S of sites, the cross block of the periodic covariance has $\operatorname{rank} C_S \le 2M\, b(S)$.
`Sealed.cross_block_rank`

38. The entanglement rank bound. For every pure matrix X on the physical labels and every set S of sites, $1+\Gamma_S^2 = C_S C_S^T$. Consequently, for chain data with the contraction condition whose periodic covariance is pure,

$$\operatorname{rank}(1+\Gamma_S^2) \le 2M\, b(S).$$

`Sealed.entanglement_rank`

39. The rank of $1+\Gamma^2$ in the normal form. If O is a real orthogonal matrix on the Majorana labels of n modes and $\Gamma = OD(\lambda)O^T$, then

$$\operatorname{rank}(1+\Gamma^2) = 2\,\#\{j : \lambda_j^2 \ne 1\}.$$

`Sealed.rank_normal_form`

40. The nullity lemma. For an invertible real matrix X on a disjoint union $\iota \sqcup \kappa$ of finite sets, the (ι,κ) block of X^{-1} has the same rank as the (ι,κ) block of X.
`Sealed.nullity_rank`

Majorana operators and covariances of states

Fock space of n fermionic modes is $\mathbb{C}^{\{0,1\}^n}$, with basis vectors $|s\rangle$ indexed by the configurations

$$s = (s_0, \dots, s_{n-1}) \in \{0,1\}^n$$

($s_j = 1$: mode j occupied); operators on it are complex matrices indexed by configurations, and $s^{(j)}$ is s with its bit j flipped. The Majorana labels of n modes are $x_0, \dots, x_{n-1}$ followed by $y_0, \dots, y_{n-1}$; a real matrix on the labels is written in blocks

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

for this order (the x-block first), and

$$D(\varepsilon) = \begin{pmatrix} 0 & \operatorname{diag}\varepsilon \\ -\operatorname{diag}\varepsilon & 0 \end{pmatrix}$$

for $\varepsilon \in \mathbb{R}^n$. A density matrix is a positive semidefinite matrix of trace 1. The covariance of a matrix ρ is the real matrix

$$\Gamma(\rho)_{ab} = \operatorname{Re} \tfrac{i}{2} \operatorname{Tr}(\rho\,[c_a, c_b]);$$

with this sign the vacuum $|0\cdots0\rangle$ has covariance

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Admissible and pure covariances are as in the first part: $\Gamma^T = -\Gamma$ with $1-\Gamma^T\Gamma$ positive semidefinite, respectively $\Gamma^T = -\Gamma$ with $\Gamma^2 = -1$. A unitary U implements a real matrix O on the labels when

$$U c_a U^* = \sum_b O_{ab} c_b$$

for every label a.

41. The index sets: the configurations of n modes are the maps from $\{0,\dots,n-1\}$ to $\{0,1\}$, and the Majorana labels of n modes are two copies of $\{0,\dots,n-1\}$, the x-labels and then the y-labels.
`Sealed.fock_labels`

42. The Jordan–Wigner Majorana operators of n modes. For every mode j and configurations s,t, the matrix entries are

$$\langle s|c_{x_j}|t\rangle = \prod_{k<j}(-1)^{s_k}$$

and

$$\langle s|c_{y_j}|t\rangle = -i\,(-1)^{s_j}\prod_{k<j}(-1)^{s_k}$$

when $t = s^{(j)}$, and 0 otherwise.
`Sealed.majorana_entries`

43. The canonical anticommutation relations: the Majorana operators are Hermitian, $c_a^* = c_a$, and

$$c_a c_b + c_b c_a = 2\delta_{ab}\,1$$

for all labels a, b.
`Sealed.majorana_car`

44. Density matrices and the covariance of a state: ρ is a density matrix exactly when it is positive semidefinite with $\operatorname{Tr}\rho = 1$; the covariance entry is

$$\Gamma(\rho)_{ab} = \operatorname{Re}\tfrac{i}{2}\operatorname{Tr}(\rho\,(c_a c_b - c_b c_a));$$

and for Hermitian ρ the number $\frac{i}{2}\operatorname{Tr}(\rho\,(c_a c_b - c_b c_a))$ is real, equal to $\Gamma(\rho)_{ab}$.
`Sealed.state_covariance`

45. The covariance of every matrix ρ is antisymmetric, and for labels $a \neq b$ its entry is

$$\Gamma(\rho)_{ab} = \operatorname{Re}\,(i\operatorname{Tr}(\rho\,c_a c_b)).$$

The covariance of a density matrix is admissible.
`Sealed.covariance_of_density`

46. Occupation-number states. For a configuration s the state $|s\rangle\langle s|$ is the matrix with the single nonzero entry 1 at (s,s). It is a density matrix, its covariance is

$$\begin{pmatrix} 0 & D_s \\ -D_s & 0 \end{pmatrix}$$

with D_s diagonal, $(D_s)_{jj} = -1$ if $s_j = 0$ and $+1$ if $s_j = 1$, and this covariance is pure. In particular the vacuum $|0\cdots 0\rangle$ has covariance

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

`Sealed.occupation_state`

47. The bond state. Let

$$\psi = (|00\rangle + i\,|11\rangle)/\sqrt{2}$$

on two modes, that is $\psi_{00} = 1/\sqrt{2}$, $\psi_{11} = i/\sqrt{2}$ and $\psi_{01} = \psi_{10} = 0$. Then $|\psi\rangle\langle\psi|$ is a density matrix, and its covariance has the nonzero entries

$$\Gamma_{x_0x_1} = -\Gamma_{x_1x_0} = 1$$

and

$$\Gamma_{y_0y_1} = -\Gamma_{y_1y_0} = -1.$$

`Sealed.bond_state`

48. Conjugation by an implementing unitary. Let U be a unitary on Fock space implementing a real matrix O. Then O is orthogonal, $OO^T = 1$, and for every density matrix ρ the matrix $U\rho U^*$ is a density matrix with covariance $O^T\Gamma(\rho)\,O$.

`Sealed.unitary_conjugation`

49. Hopping gates. For modes $j \neq k$ and $\varphi \in \mathbb{R}$ let

$$U = (\cos\varphi + \sin\varphi\, c_{x_j}c_{x_k})(\cos\varphi + \sin\varphi\, c_{y_j}c_{y_k}).$$

Then U is unitary; with $\theta = 2\varphi$ it acts on the Majorana operators by

$$Uc_{x_j}U^* = \cos\theta\, c_{x_j} - \sin\theta\, c_{x_k},$$
$$Uc_{x_k}U^* = \sin\theta\, c_{x_j} + \cos\theta\, c_{x_k},$$

the same formulas with y in place of x, and $Uc_aU^* = c_a$ for the labels a of every other mode. On two modes, with $j = 0$ and $k = 1$, the state $U|10\rangle\langle 10|U^*$ (mode 0 occupied) has covariance

$$\begin{pmatrix} 0 & X \\ -X^T & 0 \end{pmatrix}$$

with

$$X = \begin{pmatrix} \cos 4\varphi & -\sin 4\varphi \\ -\sin 4\varphi & -\cos 4\varphi \end{pmatrix}.$$

`Sealed.hopping_gate`

Quadratic Hamiltonians and ground states

For a real matrix h on the Majorana labels of n modes the quadratic Hamiltonian is

$$H_h = \tfrac{i}{4}\sum_{a,b} h_{ab}\, c_a c_b,$$

and the energy of a matrix ρ is $E_h(\rho) = \operatorname{Re}\operatorname{Tr}(\rho H_h)$. On real matrices Γ on the labels the energy form is $\frac{1}{4}\sum_{a,b} h_{ab}\Gamma_{ab}$. A normal form of h is a factorization $h = OD(\varepsilon)O^T$ with O real orthogonal and $\varepsilon \in \mathbb{R}^n$.

50. Quadratic Hamiltonians. For every real h,

$$H_h = \sum_{a,b} \tfrac{i}{4} h_{ab}\, c_a c_b$$

and $E_h(\rho) = \operatorname{Re}\operatorname{Tr}(\rho H_h)$. If $h^T = -h$, then H_h is Hermitian; for Hermitian ρ the trace $\operatorname{Tr}(\rho H_h)$ is real and equals $\frac{1}{4}\sum_{a,b} h_{ab}\Gamma(\rho)_{ab}$; and the generator identity

$$i\,[H_h, c_a] = \sum_b h_{ab}c_b$$

holds. For every h, a unitary U implementing a real matrix O satisfies $UH_hU^* = H_{O^ThO}$.

`Sealed.quadratic_hamiltonian`

51. The Hamiltonian of $D(\varepsilon)$ is diagonal in the occupation basis:

$$\langle s|H_{D(\varepsilon)}|s\rangle = \sum_j \varepsilon_j (s_j - \tfrac{1}{2})$$

for every configuration s, and all off-diagonal entries vanish.
`Sealed.block_hamiltonian`

52. Ground states in a normal form. Let $h = OD(\varepsilon)O^T$ with O orthogonal, let U be a unitary implementing O, let s^ε be the configuration with $s^\varepsilon_j = 1$ exactly when $\varepsilon_j < 0$, let $\rho_0 = U^*|s^\varepsilon\rangle\langle s^\varepsilon|U$ and

$$E_0 = -\tfrac{1}{2}\sum_j |\varepsilon_j|.$$

Then $E_h(\rho) \geq E_0$ for every density matrix ρ; ρ_0 is a density matrix with $E_h(\rho_0) = E_0$; its covariance is $OD(\eta)O^T$ with $\eta_j = 1$ if $\varepsilon_j < 0$ and $\eta_j = -1$ otherwise, and is pure. If moreover every $\varepsilon_j \neq 0$, then ρ_0 is the only density matrix with energy E_0, the energy form satisfies

$$\tfrac{1}{4}\sum_{a,b} h_{ab}\Gamma_{ab} \geq E_0$$

for admissible Γ, and for admissible Γ equality holds exactly when $\Gamma = \Gamma(\rho_0)$.
`Sealed.normal_form_ground`

53. Every quadratic Hamiltonian has a pure ground state. For every real h with $h^T = -h$ there are a special orthogonal O ($OO^T = 1$, $\det O = 1$), a vector ε with $h = OD(\varepsilon)O^T$, and a unitary U implementing O; and there is a density matrix ρ_0 with pure covariance and $E_h(\rho_0) \leq E_h(\rho)$ for every density matrix ρ.
`Sealed.ground_state_exists`

Realization

A real matrix Γ on the labels is realized by a density matrix ρ when $\Gamma(\rho) = \Gamma$. For $\lambda \in \mathbb{R}^n$ the product mixture is

$$\rho_\lambda = \sum_s \prod_j p_j(s_j)\,|s\rangle\langle s|$$

with $p_j(0) = (1 - \lambda_j)/2$ and $p_j(1) = (1 + \lambda_j)/2$.

54. Implementing unitaries exist: for every real matrix O on the labels with $OO^T = 1$ and $\det O = 1$ there is a unitary U on Fock space implementing O.
`Sealed.special_orthogonal_unitary`

55. Product mixtures. For every $\lambda \in \mathbb{R}^n$,

$$\rho_\lambda = \sum_s \prod_j p_j(s_j)\,|s\rangle\langle s|$$

has covariance $D(\lambda)$; if every $|\lambda_j| \leq 1$ it is a density matrix; and if every $|\lambda_j| = 1$ it is the occupation state $|s\rangle\langle s|$ with $s_j = 1$ exactly when $\lambda_j = 1$.
`Sealed.product_mixture`

56. Realization of covariances. If $\Gamma = OD(\varepsilon)O^T$ is admissible, O is orthogonal and U is a unitary implementing O, then $U^*\rho_\varepsilon U$ is a density matrix with covariance Γ. Every admissible Γ is the covariance of a density matrix. Every pure Γ is the covariance of a density matrix $U^*|s\rangle\langle s|U$ with U unitary and s a configuration. Two density matrices with the same pure covariance Γ are equal.

`Sealed.covariance_realization`

Thermal states and evolution

e^X is the exponential of a square matrix X. For a real matrix h on the labels the exponential state is

$$\rho_h = e^{-H_h}/\operatorname{Tr} e^{-H_h},$$

the thermal state at inverse temperature $\beta \in \mathbb{R}$ is $\rho_{\beta h}$, and the evolution operator is $U_h(t) = e^{-itH_h}$ for $t \in \mathbb{R}$. A real matrix Γ on the labels is strictly admissible when $\Gamma^T = -\Gamma$ and $1 - \Gamma^T\Gamma$ is positive definite.

57. The exponential and thermal states:

$$\rho_h = (\operatorname{Tr} e^{-H_h})^{-1} e^{-H_h},$$

and the thermal state of H_h at inverse temperature β is $\rho_{\beta h}$.

`Sealed.exponential_state_def`

58. Strict admissibility: Γ is strictly admissible exactly when $\Gamma^T = -\Gamma$ and $1 - \Gamma^T\Gamma$ is positive definite; a strictly admissible matrix is admissible; and $D(\lambda)$ is strictly admissible exactly when every $|\lambda_j| < 1$.

`Sealed.strictly_admissible`

59. Exponential states. Let $h^T = -h$. Then $\operatorname{Tr} e^{-H_h}$ is a positive real number and ρ_h is a density matrix. There are a special orthogonal O and a vector ε with $h = OD(\varepsilon)O^T$ and $\Gamma(\rho_h) = OD(-\tanh(\varepsilon/2))O^T$, the hyperbolic tangent taken entrywise. The covariance $\Gamma(\rho_h)$ is strictly admissible, and for $n \geq 1$ it is not pure.

`Sealed.exponential_state`

60. Thermal states in a normal form. Let $h = OD(\varepsilon)O^T$ with O orthogonal and let U be a unitary implementing O. Then

$$\operatorname{Tr} e^{-H_h} = \prod_j 2\cosh(\varepsilon_j/2),$$

and for every $\beta \in \mathbb{R}$ the thermal state $\rho_{\beta h}$ is a density matrix with covariance $OD(-\tanh(\beta\varepsilon/2))O^T$ and energy

$$E_h(\rho_{\beta h}) = -\tfrac{1}{2}\sum_j \varepsilon_j \tanh(\beta\varepsilon_j/2).$$

`Sealed.thermal_state`

61. Realization by exponential states. Every strictly admissible Γ is the covariance of an exponential state ρ_h with $h^T = -h$. Two exponential states $\rho_h, \rho_{h'}$ ($h^T = -h$, $h'^T = -h'$) with the same covariance are equal. If O is orthogonal, U is a unitary implementing O and every $|\eta_j| < 1$, then $h = OD(-2\operatorname{artanh}\eta)O^T$ has $\Gamma(\rho_h) = OD(\eta)O^T$. The exponential state of $h = 0$ is the maximally mixed state $2^{-n}\,1$, with covariance 0.

`Sealed.exponential_realization`

62. Time evolution. Let $h^T = -h$ and $s, t \in \mathbb{R}$. Then $U_h(t) = e^{-itH_h}$ is unitary, $U_h(t)^* = U_h(-t)$ and $U_h(s)U_h(t) = U_h(s+t)$; $U_h(t)$ implements the real matrix e^{-th}, that is

$$U_h(t) c_a U_h(t)^* = \sum_b (e^{-th})_{ab} c_b;$$

for every density matrix σ the matrix $U_h(t)\sigma U_h(t)^*$ is a density matrix with covariance $e^{th}\,\Gamma(\sigma)\,e^{-th}$; and the exponential state is stationary, $U_h(t)\rho_h U_h(t)^* = \rho_h$.
`Sealed.time_evolution`

Wick's theorem and the Gaussian class

For a complex vector v on the labels,

$$v \cdot c = \sum_a v_a c_a,$$

and

$$v^T G w = \sum_{a,b} v_a G_{ab} w_b$$

(no complex conjugation). The two-point matrix of ρ is the complex matrix $G(\rho) = 1 - i\,\Gamma(\rho)$. Lists $(w_0, \ldots, w_{k-1})$ of vectors are indexed from 0, and a hat marks an omitted factor. A Gaussian state is a density matrix whose moments obey Wick's rule in the recursive form of the first entry below.

63. The Gaussian class. A matrix ρ is Gaussian exactly when it is a density matrix and, with $G = G(\rho)$, for every vector v and every list $(w_0, \ldots, w_{k-1})$,

$$\mathrm{Tr}(\rho\,(v \cdot c)(w_0 \cdot c) \cdots (w_{k-1} \cdot c)) = \sum_{j<k} (-1)^j\,(v^T G w_j)\,\mathrm{Tr}(\rho\,(w_0 \cdot c) \cdots \widehat{(w_j \cdot c)} \cdots (w_{k-1} \cdot c)).$$

`Sealed.gaussian_iff`

64. Order two. For complex vectors v, w on the labels,

$$(v \cdot c)(w \cdot c) + (w \cdot c)(v \cdot c) = 2\,(v^T w)\,1,$$

and every density matrix ρ satisfies

$$\mathrm{Tr}(\rho\,(v \cdot c)(w \cdot c)) = v^T G(\rho)\,w.$$

`Sealed.two_point_moment`

65. The relation form of the Gaussian class. A matrix ρ is Gaussian exactly when it is a density matrix,

$$\mathrm{Tr}(\rho\,(w_0 \cdot c) \cdots (w_{k-1} \cdot c)) = 0$$

for every list of odd length k, and, with $G = G(\rho)$,

$$((G^T v) \cdot c)\,\rho = \rho\,((G v) \cdot c)$$

for every vector v.
`Sealed.gaussian_relation`

66. Examples of Gaussian states: the product mixtures ρ_λ with every $|\lambda_j| \le 1$; the occupation states $|s\rangle\langle s|$; for $h^T = -h$ the exponential state ρ_h and every thermal state $\rho_{\beta h}$, $\beta \in \mathbb{R}$. If U is a unitary implementing a real matrix O and ρ is Gaussian, then $U^*\rho U$ is Gaussian.
`Sealed.gaussian_examples`

67. Gaussian states and their covariances. Every admissible Γ is the covariance of a Gaussian state, and two Gaussian states with the same covariance are equal. Every Gaussian state ρ has the form $\rho = U^*\rho_\varepsilon U$ with U a unitary implementing an orthogonal O, every $|\varepsilon_j| \le 1$ and $\Gamma(\rho) = OD(\varepsilon)O^T$. A Gaussian state with strictly admissible covariance is an exponential state ρ_h with $h^T = -h$.
`Sealed.gaussian_covariance`

68. The Majorana monomials. For a set S of labels, c_S is the product of the c_a, $a \in S$, in the order

$$x_0 < \cdots < x_{n-1} < y_0 < \cdots < y_{n-1}$$

(so $c_\emptyset = 1$). For all sets S, T of labels, $\mathrm{Tr}(c_S^* c_T) = 2^n$ if $S = T$ and 0 otherwise; the monomials, indexed by the sets of labels, are linearly independent over $\mathbb{C}$; every complex matrix A on Fock space equals $2^{-n}\sum_S \mathrm{Tr}(Ac_S^*)\, c_S$; and two matrices A, B with $\mathrm{Tr}(Ac_S) = \mathrm{Tr}(Bc_S)$ for every S are equal.
`Sealed.majorana_monomials`

Partial trace and contraction of Gaussian states, the chain state

A configuration of $n+m$ modes is written (s,u), with $s \in \{0,1\}^n$ its first n bits (the retained modes) and $u \in \{0,1\}^m$ its last m bits (the eliminated modes). The retained Majorana labels are x_j, y_j for $j < n$ and the eliminated ones x_{n+j}, y_{n+j} for $j < m$; for a real matrix Γ on the labels of $n+m$ modes, A, B, C, D are its retained–retained, retained–eliminated, eliminated–retained and eliminated–eliminated blocks, each with the x-labels before the y-labels. Tr_E and Tr_R are the partial traces over the eliminated and over the retained modes, and $X \otimes Y$ is the tensor product of a matrix X on n modes with a matrix Y on m modes. The contraction of a matrix ρ on $n+m$ modes with a matrix σ on m modes has normalization $Z = \mathrm{Tr}(\rho\,(1 \otimes \sigma))$ and is the matrix $Z^{-1}\,\mathrm{Tr}_E(\rho\,(1 \otimes \sigma))$ on n modes.

69. The partial traces and the tensor product: for matrices M on $n+m$ modes, X on n modes and Y on m modes and for all configurations,

$$\mathrm{Tr}_E(M)_{st} = \sum_w M_{(s,w),(t,w)},$$

$$\mathrm{Tr}_R(M)_{uv} = \sum_w M_{(w,u),(w,v)}$$

and

$$(X \otimes Y)_{(s,u),(t,v)} = X_{st}\, Y_{uv}.$$

`Sealed.partial_trace`

70. Marginals. For every matrix ρ on $n+m$ modes, the covariance of $\mathrm{Tr}_E\rho$ is the retained block A of $\Gamma(\rho)$ and the covariance of $\mathrm{Tr}_R\rho$ is the eliminated block D. If ρ is a density matrix, so are both marginals; if ρ is Gaussian, so are both marginals.
`Sealed.gaussian_marginal`

71. Contraction of Gaussian states. For positive semidefinite ρ on $n+m$ modes and σ on m modes with $Z = \mathrm{Tr}(\rho\,(1\otimes\sigma)) \neq 0$, the contraction $Z^{-1}\,\mathrm{Tr}_E(\rho\,(1\otimes\sigma))$ is a density matrix. Let ρ and σ be Gaussian, with A, B, C, D the blocks of $\Gamma(\rho)$ and $\beta = \Gamma(\sigma)$, and let $1 - D\beta$ be invertible. Then Z is a positive real number, the contraction is Gaussian, and its covariance is $A + B\beta(1 - D\beta)^{-1}C$.

`Sealed.gaussian_contraction`

72. The chain state. Let $\Gamma_0, \ldots, \Gamma_{N-1}$ be admissible covariances of single sites with P physical and $2M$ virtual modes, let Γ_{in}, β_N and Γ_N be the periodic input, bond and periodic covariances of the chain built from them (defined in the first part), and let the invertibility condition of the contraction of Γ_{in} with β_N (also defined there) hold. Number the modes of the chain as follows: physical mode c of site i is mode $iP + c$; the NM virtual modes L follow, (i, a) being mode $NP+iM+a$; then the NM virtual modes L', (i, b) being mode $NP+NM+iM+b$. On the $2NM$ virtual modes alone, L is numbered $(i, a) \mapsto iM + a$ and L' is numbered $(i, b) \mapsto NM + iM + b$. Then there are Gaussian states ρ on all $NP+2NM$ modes and σ on the $2NM$ virtual modes whose covariances, read in this numbering, are Γ_{in} and β_N. For any such ρ and σ, the normalization $Z = \mathrm{Tr}(\rho\,(1\otimes\sigma))$ is a positive real number, the contraction

$$\tau = Z^{-1}\,\mathrm{Tr}_E(\rho\,(1\otimes\sigma))$$

on the NP physical modes is Gaussian, its covariance, read in the numbering $iP + c$, is Γ_N, and if every Γ_i is pure then $\tau^2 = \tau$.

`Sealed.chain_state`

Purity and the Kitaev ground state

For a pure Γ on the labels of n modes, $H_{-\Gamma}$ is the quadratic Hamiltonian of $-\Gamma$. The Kitaev chain has N sites with one mode each; mode i has the labels x_i, y_i, and $i+1$ is taken modulo N. The Kitaev covariance K is the real matrix on these labels with $K_{y_i x_{i+1}} = 1$, $K_{x_{i+1} y_i} = -1$ and all other entries 0.

73. Gaussian projections: a Gaussian state ρ satisfies $\rho^2 = \rho$ exactly when its covariance is pure, and then $\rho = |\psi\rangle\langle\psi|$ for a vector ψ.

`Sealed.gaussian_projection`

74. Contraction of pure Gaussian states. Let ρ on $n+m$ modes and σ on m modes be Gaussian with pure covariances, and let $1 - D\beta$ be invertible, with D the eliminated block of $\Gamma(\rho)$ and $\beta = \Gamma(\sigma)$. Then the contraction

$$\tau = Z^{-1}\,\mathrm{Tr}_E(\rho\,(1\otimes\sigma))$$

satisfies $\tau^2 = \tau$ and its covariance is pure.

`Sealed.pure_contracted_state`

75. Ground states of pure covariances. For every pure Γ on the labels of n modes there is a Gaussian state ρ_0 with covariance Γ such that $E_{-\Gamma}(\rho_0) = -n/2$, $E_{-\Gamma}(\rho) \geq -n/2$ for every density matrix ρ, and every density matrix with $E_{-\Gamma}(\rho) = -n/2$ equals ρ_0; on covariances,

$$-\tfrac{1}{4}\sum_{a,b} \Gamma_{ab}\Gamma'_{ab} \geq -n/2$$

for every admissible Γ', with equality exactly when $\Gamma' = \Gamma$.

`Sealed.pure_ground_state`

76. The Kitaev covariance and Hamiltonian. On the labels of the N modes of the Kitaev chain, for all i, j: $K_{x_i x_j} = K_{y_i y_j} = 0$, $K_{y_i x_j} = 1$ if $j = i+1$ and 0 otherwise, and $K_{x_j y_i} = -1$ if $j = i+1$ and 0 otherwise. The matrix K is pure, and

$$H_{-K} = \tfrac{i}{2} \sum_i c_{x_{i+1}} c_{y_i}.$$

`Sealed.kitaev_hamiltonian`

77. The Kitaev ground state as a chain state. Let $N \geq 1$, and take the chain of N sites with $P = 1$ physical and $2M = 2$ virtual modes per site, every site carrying the Kitaev local covariance (defined in the first part), with the numbering of the modes of the chain-state entry. Then there are Gaussian states ρ and σ whose covariances, in this numbering, are the periodic input and bond covariances. For any such ρ and σ the contraction

$$\rho_K = Z^{-1} \operatorname{Tr}_E(\rho\,(1 \otimes \sigma))$$

is Gaussian with covariance K, satisfies $\rho_K^2 = \rho_K$, has energy $E_{-K}(\rho_K) = -N/2$, and is the unique ground state of H_{-K}: $E_{-K}(\rho') \geq -N/2$ for every density matrix ρ', and every density matrix with $E_{-K}(\rho') = -N/2$ equals ρ_K.
`Sealed.kitaev_ground_state`

The gap

For a pure Γ the Hamiltonian $H_{-\Gamma}$ has ground energy $-n/2$ and a unique ground state (previous section). The statements below bound the energy of density matrices orthogonal to the ground state, orthogonality of ρ and ρ_0 meaning $\operatorname{Tr}(\rho\,\rho_0) = 0$.

78. The gap of $H_{-\Gamma}$. Let Γ be pure and let ρ_0 be a Gaussian state with covariance Γ. Every density matrix ρ with $\operatorname{Tr}(\rho\,\rho_0) = 0$ has $E_{-\Gamma}(\rho) \geq -n/2 + 1$, and for $n \geq 1$ some density matrix ρ' with $\operatorname{Tr}(\rho'\rho_0) = 0$ has $E_{-\Gamma}(\rho') = -n/2 + 1$.
`Sealed.pure_gap`

79. The gap of the Kitaev chain. Let $N \geq 1$ and let ρ_K be a density matrix on the N modes of the Kitaev chain with $E_{-K}(\rho_K) = -N/2$. Every density matrix ρ with $\operatorname{Tr}(\rho\,\rho_K) = 0$ has $E_{-K}(\rho) \geq -N/2 + 1$, and some density matrix ρ' with $\operatorname{Tr}(\rho'\rho_K) = 0$ has $E_{-K}(\rho') = -N/2 + 1$.
`Sealed.kitaev_gap`